

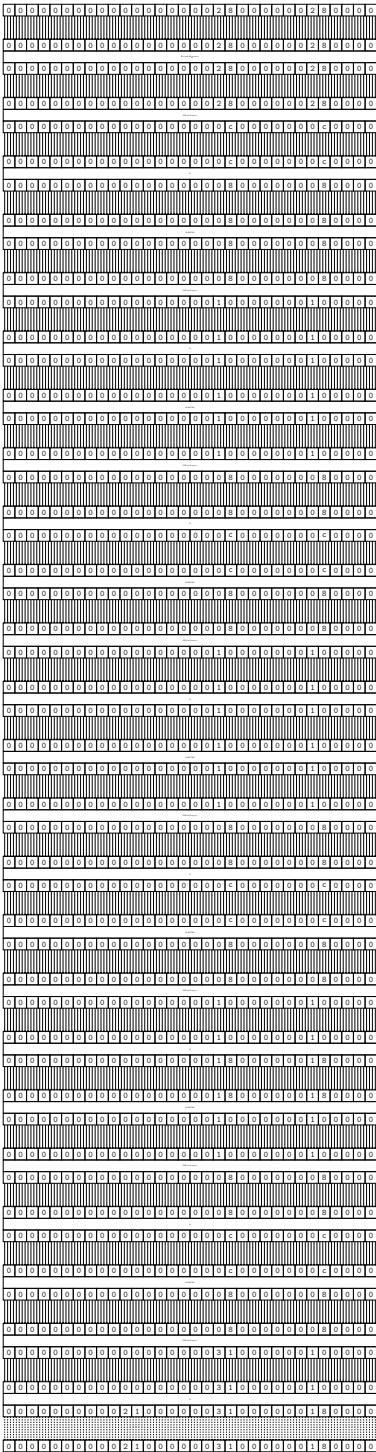
Bitwise linear trail through **QALQAN** found by SAT

CiVerLy

August 21, 2026

1 QALQAN

Maximal linear correlation: 2^{-78}



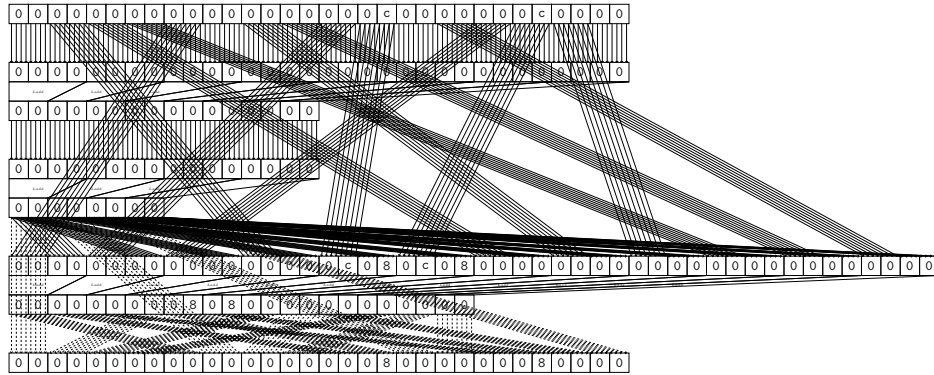
2 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	8	0	0	0	0	0	0	2	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	8	0	0	0	0	0	0	2	8	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	c	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	c	0	0	0	0	

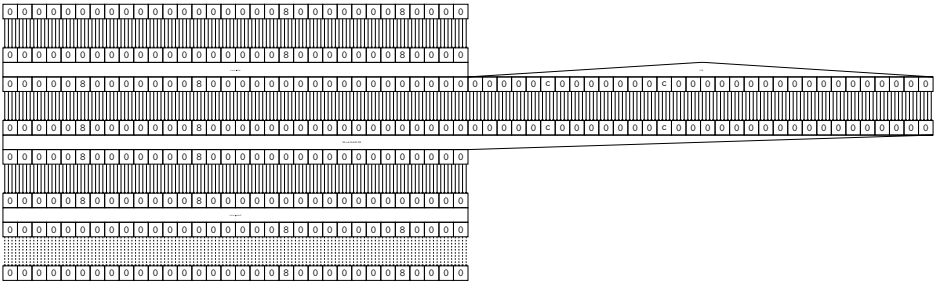
3 L

Maximal linear correlation: $2^{-2.0}$



4 Add128

Maximal linear correlation: $2^{-2.0}$

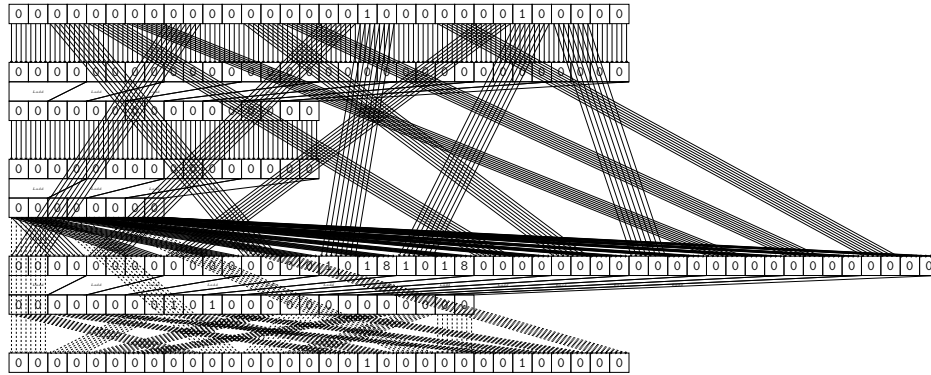


5 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

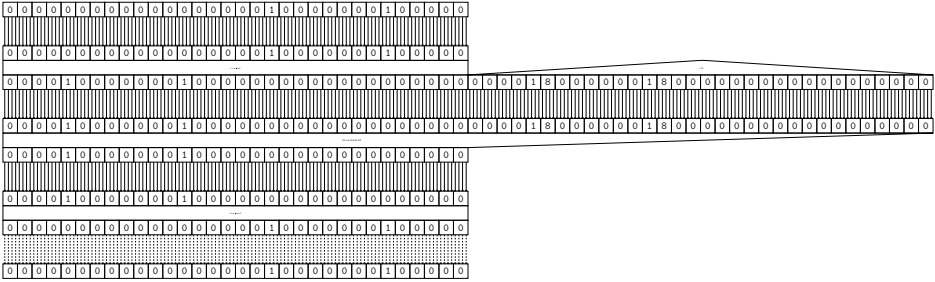
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0

6 L

Maximal linear correlation: $2^{-2.0}$ 

7 Add128

Maximal linear correlation: $2^{-2.0}$



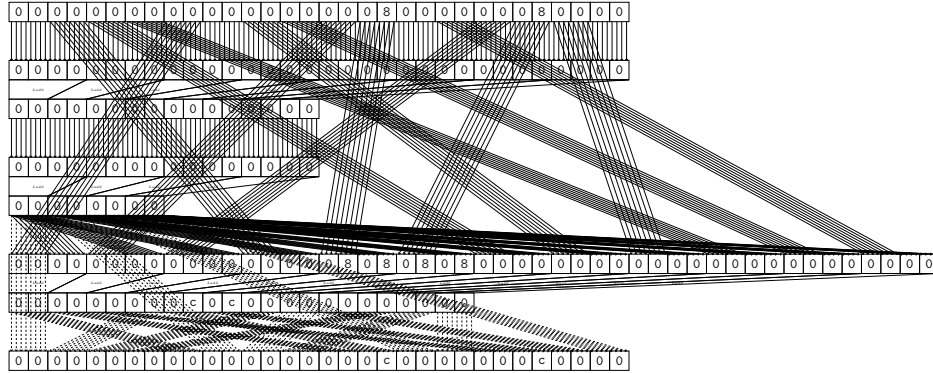
8 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0

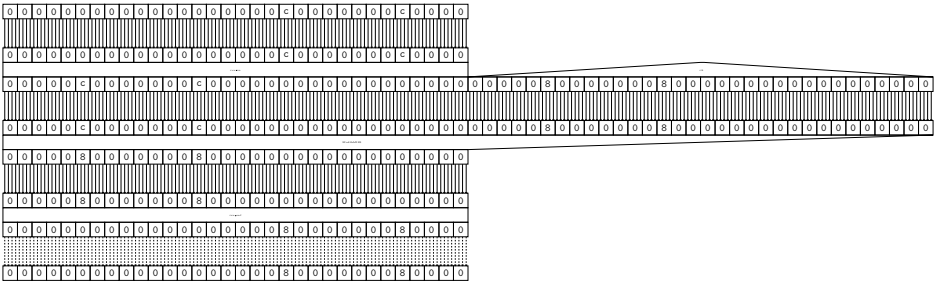
9 L

Maximal linear correlation: $2^{-2.0}$



10 Add128

Maximal linear correlation: $2^{-2.0}$



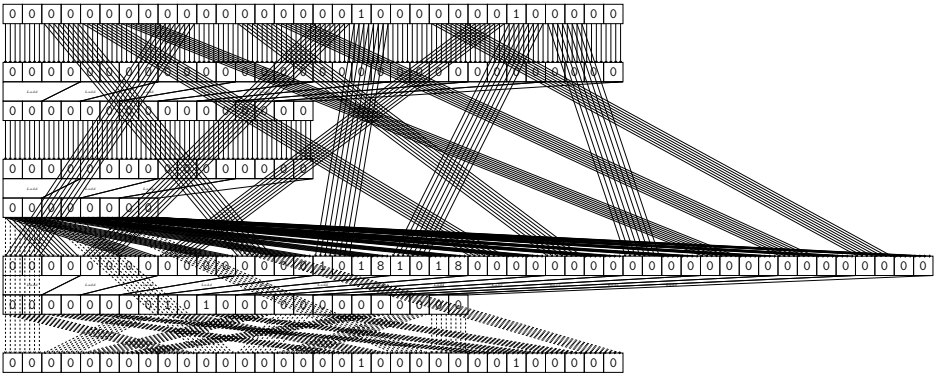
11 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	

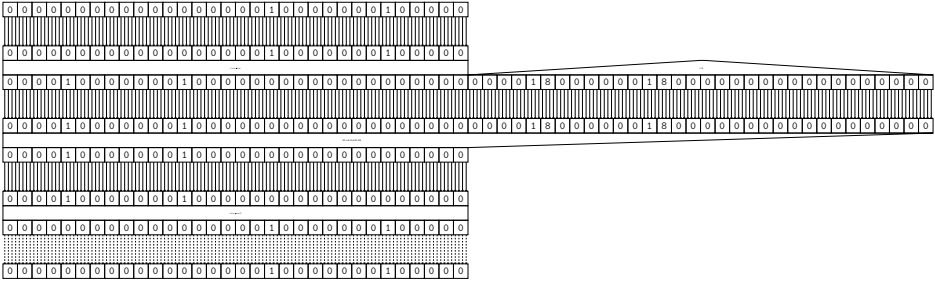
12 L

Maximal linear correlation: $2^{-2.0}$



13 Add128

Maximal linear correlation: $2^{-2.0}$

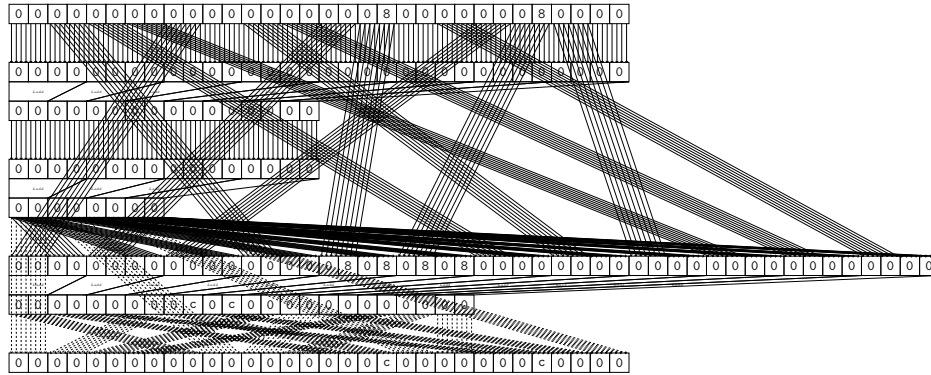


14 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

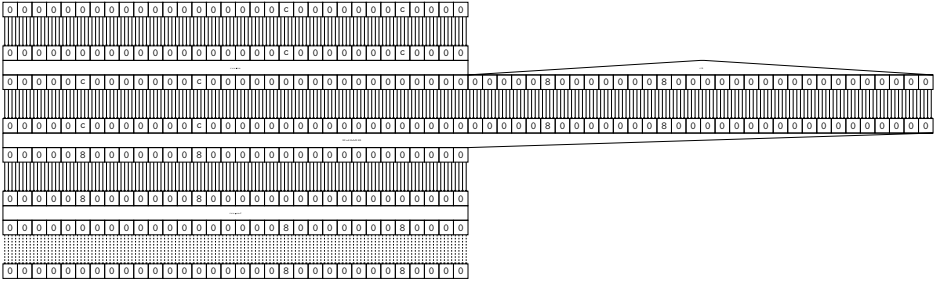
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0

15 L

Maximal linear correlation: $2^{-2.0}$ 

16 Add128

Maximal linear correlation: $2^{-2.0}$



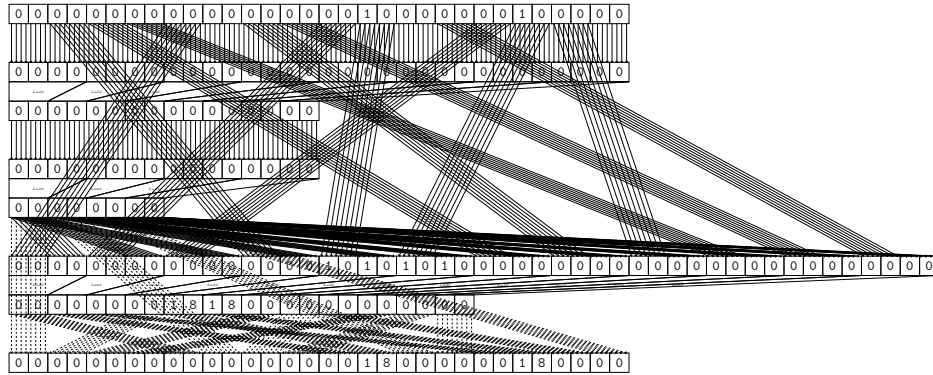
17 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

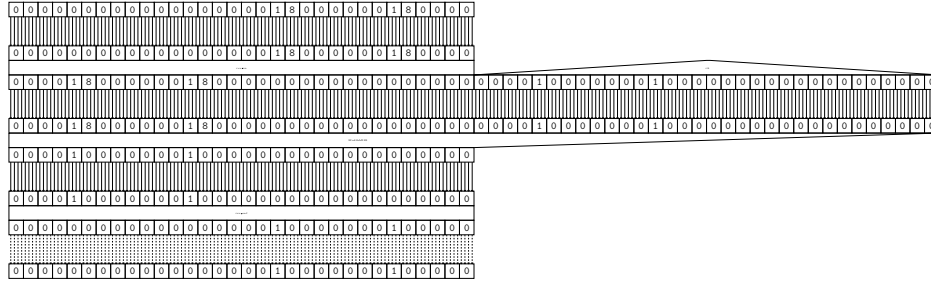
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	

18 L

Maximal linear correlation: $2^{-2.0}$



19 Add128

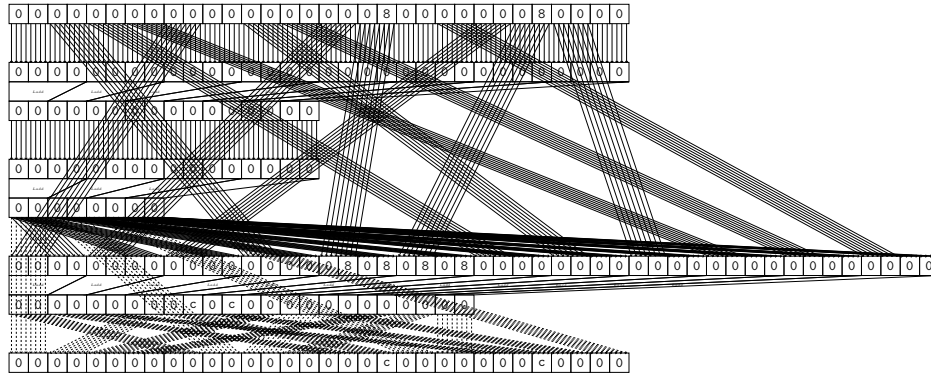
Maximal linear correlation: $2^{-2.0}$ 

20 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

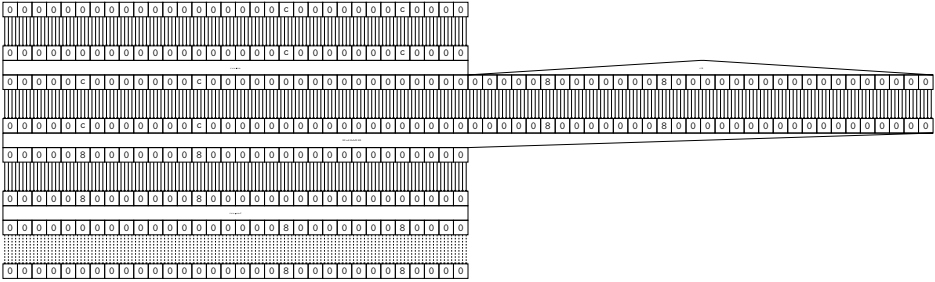
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0

21 L

Maximal linear correlation: $2^{-2.0}$ 

22 Add128

Maximal linear correlation: $2^{-2.0}$



23 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	1	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	1	0	0	0	0	0	1	0	0	0	0	0

24 L

Maximal linear correlation: $2^{-2.0}$ 